Machine Learning in Python

Neural Networks II: Deep Learning, Transformers, and GenAI

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Outline

Why CNNs?

The Problem with MLPs for Images

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- Spatial Structure Lost: A nose is defined by being *above* a mouth, not just by pixel values.
- Parameter Explosion: A 1000×1000 image connected to 100 neurons requires 100 million weights.

The Convolution Operation

Filters and Feature Maps

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- It slides (convolves) over the image, computing dot products.
- This creates a Feature Map showing where patterns appear.
- Early filters detect edges; deeper filters detect complex shapes.

Controlling Dimensions

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- Formula: $O = \frac{I-K+2P}{S} + 1$

Pooling Layers

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- Typical: 2×2 pool with stride 2 halves dimensions.

Famous Architectures

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- Skip connections let gradients "skip" layers to avoid vanishing.
- Enables training networks with 100+ layers.

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- We need models that handle variable-length sequences.
- We need models that "remember" context.

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RNNs and Their Limitations

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- LSTMs improved memory with "gates," but remained slow.

The Attention Mechanism

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- "I poured cereal into the..." $\rightarrow$ Focus on "cereal" and "poured" $\rightarrow$ Predict: "bowl".

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In 2014, researchers asked: *Why squash everything into one tiny memory vector?* Attention lets the model "look back" at any word:

- When predicting, assign a "weight" to every previous word.
- "I poured cereal into the..." $\rightarrow$ Focus on "cereal" and "poured" $\rightarrow$ Predict: "bowl".
- No more sequential bottleneck for context.

The Transformer (2017)

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The paper proved we don't need recurrence at all:

Feature	RNN/LSTM	Transformer
Processing	Sequential (Slow)	Parallel (Fast)
Context	Local (Nearby words)	Global (All words see all words)
Distance	Linearly far	Constant (1 step away)

Transformer Architecture

Key Components

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- Positional Encoding: Injects word order since parallel processing loses it.

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- Today's frontier: Claude, GPT-4, Gemini, Llama-3.

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- Inference: Using the model. Feed prompt, generate response.

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- GPT-4: 128k tokens. Claude: 200k tokens.

Prompt Engineering

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- Chain of Thought: "Think step by step." Dramatically improves reasoning.

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- It samples from this distribution to generate text.

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  3. Generate: LLM answers using provided context.

Diffusion Models

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- Result: Clear image emerges from noise.

Ethical Considerations

Bias, Hallucination, and Copyright

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- Mitigation: Data curation, RLHF, human oversight.

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4. Generative AI: Text (next-token), Images (diffusion). RAG grounds LLMs in facts.
5. Ethics: AI reflects training data. Requires human oversight and responsibility.